

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		incorrect since other cations form soluble salts with these anions (1) carry out flame test – if flame turns yellow cation is sodium (1)	1		1	2		1
	(b)	(i)		A is simple molecular (1) D is ionic (1) B is probably giant molecular but could be ionic since not all ionic substances are soluble (1) C is probably metallic but could be giant molecular (graphite) since graphite conducts electricity (1)	2		2	4		
		(ii)		B - use higher temperatures to melt the solid and test the molten substance for electrical conductance (1) if B conducts then ionic / if B does not conduct then giant molecular (1) OR C - add acid (and warm) and see if effervescence forms (1) if C effervesces then metal / if C does not effervesce then graphite (1) credit sensible alternative physical tests e.g. hardness of B , malleability of C		1	1	2		2
		(iii)		can be difficult to get powdered metal to conduct electricity / powdered metal not shiny / not sonorous / not malleable		1		1		1
				Question 11 total	5	4	6	15	0	10